

Inverse Matrix

In Matrix, the commutative law for multiplication is not valid.

Meaning $AB \neq BA$.

So, the valid matrix arithmetic are:

$$\boxed{A+B=B+A}$$

$$\boxed{A+(B+C)=(A+B)+C}$$

$$\boxed{A(BC)=(AB)C}$$

$$\boxed{A(B+C)=AB+AC}$$

$$\boxed{(B+C)A=BA+CA}$$

⊗ If A is an $n \times n$ matrix, then, $AI_n = A$ and $I_n A = A$

⊗ If R is the RREF of an $n \times n$ matrix A , then either R has a row of zeros or R is the identity matrix I_n .

$$\otimes AA^{-1} = I$$

→ If A is a square matrix, and if B is a matrix of the same size can be found such that $AB=BA=I$, then A is said to be invertible and B is called an inverse of A .

If no such matrix B can be found, then A is said to be singular.

$$\otimes A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\otimes (AB)^{-1} = B^{-1}A^{-1}. \text{ Also } (ABC)^{-1} = C^{-1}B^{-1}A^{-1}$$

$$\otimes A^0 = I \text{ and } A^{-n} = (A^{-1})^n$$

$$\otimes A^R A^S = A^{R+S}, (A^R)^S = A^{RS}, (kA)^{-1} = \frac{1}{k} A^{-1}$$

⊗ If A is invertible, the one and only solution

$$\text{to } Ax=b \text{ is } x=A^{-1}b$$

⊗ If there is a nonzero vector z such that $Az=0$.

Then A has dependent columns. It cannot have an inverse.

No matrix can bring 0 back to z .

So, a square matrix is invertible if and only if its columns are independent.

(Try explaining why it is not invertible if

(Try explaining why it is not invertible if a matrix has dependent columns.)

Elimination Matrices

Consider an example:

$$A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix}, \text{ then } E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$$

$$\text{So, } EA = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 4 & 4 & 10 & 9 \end{bmatrix} \quad \hookrightarrow \text{Here, } R_3 \leftarrow 3R_1 + R_3$$

$\hookrightarrow$ So, some row operations on I will produce E .

E can be obtained,

$\rightarrow$ Multiplying row i by c ($c \neq 0$)

$\rightarrow$ Interchanging rows i and j

$\rightarrow$ Add c times row i to row j .

Basically more formal language of Gaussian Elimination

So, for a 3×3 matrices,

Example 1

$$A = \begin{bmatrix} 3 & 1 & 0 \\ -3 & 1 & 1 \\ 6 & 8 & 4 \end{bmatrix}. \text{ Find } E \text{ and } U$$

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 1 & 0 \\ -3 & 1 & 1 \\ 6 & 8 & 4 \end{bmatrix}$$

$$\hookrightarrow A' = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 8 & 4 \end{bmatrix}$$

$$\hookrightarrow A' = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 1 \\ - & - & - \end{bmatrix}$$

$$\hookrightarrow A' = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 6 & 4 \end{bmatrix} \rightarrow R_3 = R_3 - 3R_2$$

$$\hookrightarrow A' = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow U$$

Sol. $E = E_{21} E_{31} E_{32}$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -2 & -3 & 1 \end{bmatrix}$$

$\therefore U = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Example 2.2(3):

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 4 & 0 & 1 \\ -2 & 2 & 0 \end{bmatrix}, \text{ Find } E_{32} E_{31} E_{21} A, E \text{ and } U.$$

Sol: $A = \begin{bmatrix} 1 & 1 & 0 \\ 4 & 0 & 1 \\ -2 & 2 & 0 \end{bmatrix} \rightarrow R_2 = R_2 - 4R_1$

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_{21}A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -4 & 1 \\ -2 & 2 & 0 \end{bmatrix} \rightarrow R_3 = R_3 + 2R_2$$

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -4 & 1 \\ 2 & 0 & 1 \end{bmatrix}$$

$$E_{31} E_{21} A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -4 & 1 \\ 0 & -6 & 1 \end{bmatrix} \rightarrow R_3 = R_3 - 2R_2$$

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -4 & 1 \\ 0 & -2 & 1 \end{bmatrix}$$

$\downarrow$

$$E_{32}E_{31}E_{21}A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

$$\Rightarrow U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

Here, $E_{32}E_{31}E_{21} = E = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 10 & 2 & 1 \end{bmatrix}$

$$\hookrightarrow E^{-1} = L = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ -2 & 2 & 1 \end{bmatrix}$$

So, L is the inverse of E .

Finding A^{-1} using Gauss Jordan Elimination.

If A is a matrix, you can find A^{-1} using Gauss Jordan Elimination.

Keep the form in $[A \ I]$.

$\hookrightarrow$ Transform A is Reduced Row Echelon Form.

Eventually you will get $[I \ A^{-1}]$.

Example:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right] \quad R_3 = R_3 - R_1$$

$$= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right] \quad R_2 = R_2 - 2R_1$$

$$= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right] \quad R_3 = R_3 + 2R_2$$

$$\rightarrow \text{REF} \quad \hookrightarrow \text{Gauss Elimination}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \quad R_2 = R_2 + 3R_3$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & -25 & 0 & 6 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \quad R_1 = R_1 - 2R_2$$

$$\rightarrow \text{RREF} \quad \hookrightarrow \text{Gauss Jordan Elimination}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 0 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \quad R_1 = R_1 - 3R_3$$

$$= \left[\begin{array}{cc} I & A^{-1} \end{array} \right]$$

$$\therefore A^{-1} = \begin{bmatrix} -40 & 10 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

⊗ If we find a row full of zeros after applying some row operations then it is not invertible.

Elementary Matrix

Elementary Matrices

An **elementary matrix** is a square matrix obtained by performing a **single elementary row operation** on the identity matrix. Since elementary row operations are invertible, elementary matrices are always invertible. There are three types of elementary matrices corresponding to the three types of elementary row operations:

1. **Row swapping (E_1)**: Interchanging two rows.
2. **Row scaling (E_2)**: Multiplying a row by a nonzero scalar.
3. **Row addition (E_3)**: Adding a multiple of one row to another.

For example:

Row Swapping $E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightsquigarrow$ Applying E_1 to a matrix swaps the first and second rows.

Row Scaling $E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightsquigarrow$ Applying E_2 to a matrix multiplies the second row by 3

Row Addition $E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \rightsquigarrow$ Applying E_3 to a matrix adds two times the first row to the third row.

Difference Between Elementary Matrix and Elimination Matrix

1. Elementary Matrix:

- Any matrix that represents a **single** row operation.
- Covers **all three types** of row operations (swap, scale, and row addition).
- Always **invertible**.

2. Elimination Matrix:

- A **special type** of elementary matrix used specifically for converting a matrix into an upper triangular form (Gaussian elimination).
- Only **adds multiples of one row to another** (Type 3 elementary matrices).
- **Never involves row swaps or row scaling**.
- Always **lower triangular**.

How Elimination Matrices are a Subset of Elementary Matrices

- Elimination matrices are a **type of elementary matrix** that performs only row addition.
- While elementary matrices include all three row operations (swap, scale, add), **elimination matrices only use row addition** (Type 3).
- Every elimination matrix is an elementary matrix, but **not every elementary matrix is an elimination matrix**.

Elimination matrices are a **subset** of elementary matrices, specifically those used in Gaussian elimination.



For row swapping and row scaling property, any matrix can be written as A

$$A = E_m \dots E_3 E_2 E_1$$

But if E were strictly elimination matrix:

$$A = E_m \dots E_3 E_2 E_1 U$$